

Introduction

We analyze how happiness relates to time, GDP per capita and life expectancy for different countries and continents using the cantril_ladder and countries datasets.

Methods

Data transformations. We scaled all continuous explanatory variables (time in years, GDP per capita and life expectancy) by setting mean to 0 and variance to 1.

Model. We fitted a hierarchical Bayesian with 3 levels, namely the bottom (country) level, where we used linear regression with normal errors, the middle (continent) level and the upper (global) level. It is specified as such:

Data : $i = 1, \dots, N$,
countries : $c \in \{1, \dots, C\}$,
continents : $k \in \{1, \dots, K\}$

Likelihood:

$y_i \sim \mathcal{N}(\mu_i, \sigma_y)$
 $\mu_i = \alpha_c + \beta_k^{(t)} \text{year}_i + \beta_k^{(g)} \text{gdp}_i + \beta_k^{(l)} \text{life}_i$

Country level:

$\alpha_c \sim \mathcal{N}(\alpha_0, \sigma_{\alpha_c})$

Continent level:

$\alpha_k \sim \mathcal{N}(\alpha_0, \sigma_{\alpha_k})$
 $\beta_k^{(t)} \sim \mathcal{N}(\beta_0^{(t)}, \sigma_{\beta_t})$
 $\beta_k^{(g)} \sim \mathcal{N}(\beta_0^{(g)}, \sigma_{\beta_g})$
 $\beta_k^{(l)} \sim \mathcal{N}(\beta_0^{(l)}, \sigma_{\beta_l})$

Global priors:

$\alpha_0 \sim \mathcal{N}(5, 2)$
 $\beta_0^{(t)}, \beta_0^{(g)}, \beta_0^{(l)} \sim \mathcal{N}(0, 1)$

Scale priors:

$\sigma_y, \sigma_{\alpha_c}, \sigma_{\alpha_k}, \sigma_{\beta_t}, \sigma_{\beta_g}, \sigma_{\beta_l} \sim \text{Exponential}(1)$

There is a trick to the beta parameters, namely they are specified only down to the continent level, which means that trends in time, life expectancy and GDP per capita will be shared across all countries in the same continent. We did that instead of regularization, because there are few (around 14) data points for each country. Despite that, countries still have their own intercept, which means that we learn the base happiness score for each country separately.

Diagnostics. We ran 4 chains with 2000 warm-up and 2000 accepted samples each, totaling 8000 usable samples. Diagnostics showed that everything was OK.

Prior checking. We were mainly interested in the global intercept parameter α_0 , because it determined the output distributions the most. We set its mean to 5 (middle happiness score) and variance to 2, which produced varied enough datasets, like in Figure 1.

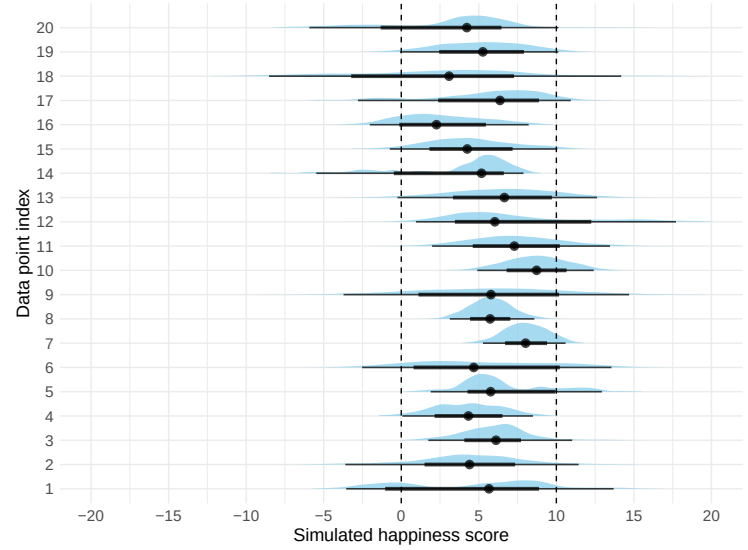


Figure 1. Prior checking. Prior predictive distributions for first 20 data points.

Posterior checking. To see if the sampled parameters can generate a dataset with a similar distribution to the original data, we compared the quartiles of the original dataset with the quartiles of 1000 replicate datasets and showed the relatively small differences in Figure 2.

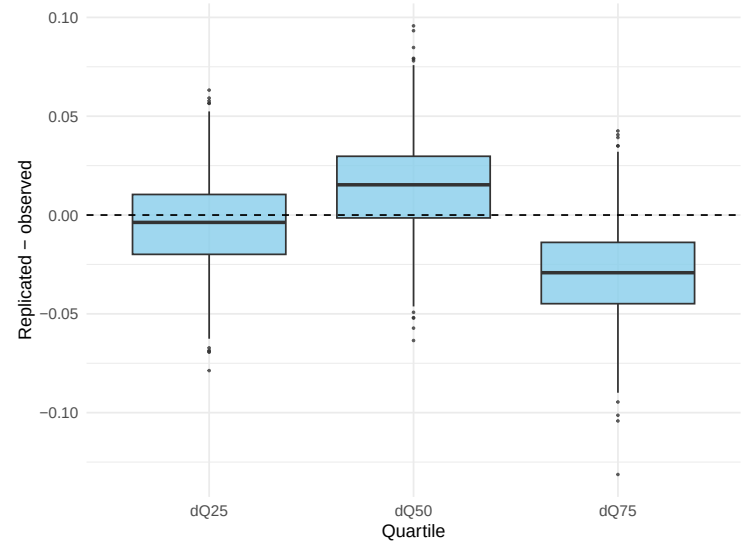


Figure 2. Posterior checking. Quartile discrepancies between original and replicate data using 1000 replicate datasets.

Results

Q1: Is happiness globally positively correlated with the year (is it increasing year-by-year)? To unscale the β_t parameter, we divided its values with the standard deviation of the time variable. The posterior for the time slope then follows an almost normal distribution with mean 0.003 and 95 % HDI [-0.029, 0.029]. That means that there is almost no gain in happiness over the years, with $P(\beta_t > 0) \approx 0.608$.

Q2: Are humans nowadays (2025) happy (score higher than 5) from the global perspective? The distribution of predicted happiness scores for 2025 showed the mean at 5.49 with 95 % HDI being [4.90, 6.11]. So the probability $P(\text{happiness} > 5) \approx 95.2\%$ showed strong evidence to support that claim.

Q3: What is the probability that Slovenia is nowadays (2025) ranked in the top 20 countries happiness wise? The probability of Slovenia being ranked top 20 is $\approx 0.000 \pm 0.000$. The posterior distribution is actually far from rank 20, the mean being around 100 as shown in Figure 3.

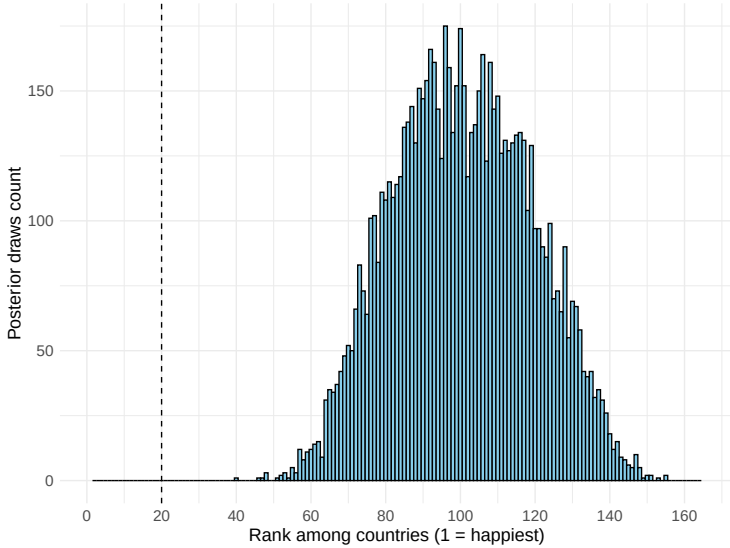


Figure 3. Posterior distribution of Slovenia's happiness rank in 2025 among 165 countries.

Q4: Are there any major differences between the correlation of health and wealth when it comes to continents? For calculating the Pearson correlation coefficient we used the continent-level slopes of the log_GDP and life_expectancy in the following formula:

$$\beta = r \cdot \frac{\rho_Y}{\rho_X}$$

$$r = \beta \cdot \frac{\rho_X}{\rho_Y} \quad (\text{due to scaling: } \rho_X = 1)$$

$$r = \frac{\beta}{\rho_Y},$$

where r is the Pearson correlation coefficient and ρ_X and ρ_Y are the standard deviations of the explanatory variable and target variable, respectively. To avoid stability issues for less represented continents (like Oceania), we used the global standard deviation instead of continent specific deviations for ρ_Y . The results are shown in Figure 4. Continent with the highest correlation in health is North America and the continent with the highest correlation in wealth is Europe. We also notice that for most continents wealth is much more linearly correlated to happiness than health.

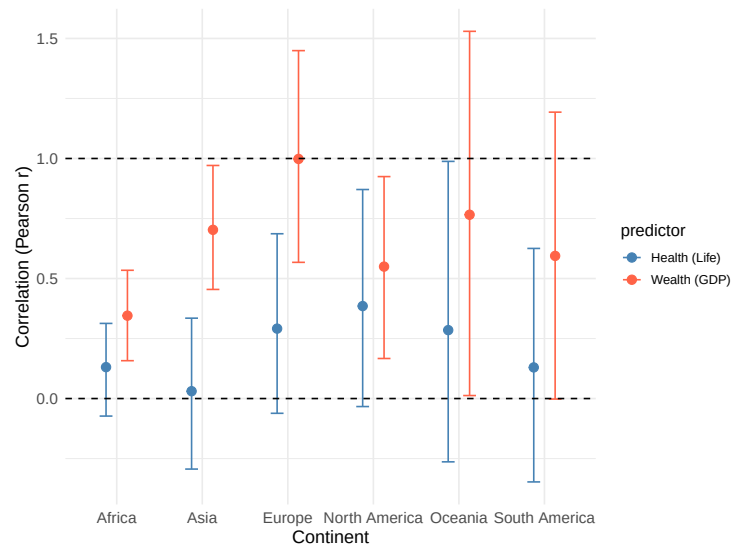


Figure 4. Continent-level correlations of health and wealth with happiness. The dots represent means and the shadows represent 95 % HDI.

Discussion

This analysis investigated global and continent-level patterns in happiness using a three-level hierarchical Bayesian model. Overall, the model demonstrated good internal consistency: diagnostics showed stable sampling, prior predictive checks produced plausible datasets, and posterior predictive checks indicated that the fitted model can reproduce key distributional features of the observed data.

Regarding temporal trends, the results provide little evidence that global happiness has increased over time. Although the posterior probability $P(\beta_t > 0) \approx 0.61$ slightly favors a positive trend, the 95% HDI for the yearly effect includes zero and is centered very close to it. Substantively, this suggests that any year-by-year change in happiness is small relative to unexplained variability.

From a global perspective, predicted happiness levels in 2025 are clearly above the neutral midpoint of the scale. The posterior probability that the global mean happiness exceeds 5 is high, indicating strong evidence for overall positive well-being. However, this result should be interpreted cautiously: the model estimates an average across heterogeneous countries, and the hierarchical structure necessarily smooths extreme national deviations toward continent-level means.

The analysis of Slovenia's rank highlights a limitation of the model's resolution at the country level. The posterior assigns essentially zero probability to Slovenia being among the top 20 happiest countries.

Differences between continents in the relationship between happiness and both wealth and health were assessed via continent-level regression slopes transformed into correlations. While this approach yields interpretable summaries, it relies on the linear model structure and uses the global standard deviation instead of continent specific ones, which may introduce bias.